18. When 15.68 litres of a gas mixture of methane and propane are fully combusted at 0^0 C and 1 atmosphere, 32 litres of oxygen at the same temperature and pressure are consumed. The amount of heat of released from this combustion in kJ is (ΔH_C (CH_4) = – 890 kJ mol^{-1} and ΔH_C (C_3H_8) = – 2220 kJ mol^{-1})

a) – 889 kJ b) – 1390 kJ c) – 3180 kJ d) – 632.68 kJ

19. The bond dissociation energy of methane and ethane are 360 kJ mol^{-1} and 620 kJ mol^{-1} respectively. Then, the bond dissociation energy of C-C bond is

a) 170 kJ mol^{-1} b) 50 kJ mol^{-1} c) 80 kJ mol^{-1} d) 220 kJ mol^{-1}

20. The correct thermodynamic conditions for the spontaneous reaction at all temperature is (NEET Phase - I)

a) $\Delta H < 0$ and $\Delta S > 0$

b) $\Delta H < 0$ and $\Delta S < 0$

c) $\Delta H > 0$ and $\Delta S = 0$

d) $\Delta H > 0$ and $\Delta S > 0$

21. The temperature of the system, decreases in an _____

a) Isothermal expansion b) Isothermal Compression c) adiabatic expansion

d) adiabatic compression

22. In an isothermal reversible compression of an ideal gas the sign of q, ΔS and w are respectively

a) +, –, – b) –, +, – c) +, –, + d) –, –, +

23. Molar heat of vapourisation of a liquid is 4.8 kJ mol^{-1}. If the entropy change is 16 J mol^{-1} K^{-1}, the boiling point of the liquid is

a) 323 K b) 27^0 C c) 164 K d) 0.3 K

24. ΔS is expected to be maximum for the reaction

a) $Ca(S) + ½\ O_2(g) \rightarrow CaO(S)$

b) $C(S) + O_2(g) \rightarrow CO_2(g)$

c) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

d) $CaCO_3(S) \rightarrow CaO(S) + CO_2(g)$

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